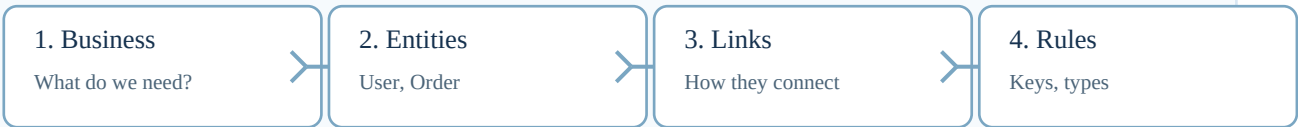


Database Modelling: Simple First Steps

A one-page guide for understanding how to design a database from the beginning.



Simple flow: understand the process before creating tables.

What to do first

- Understand the business problem. Ask: what is the system trying to do?
- List the main things the system cares about. These become tables, for example: Users, Products, Orders.
- Write the details for each thing. These become columns, for example: user_name, email, created_at.
- Show how the tables connect. Example: one customer can have many orders.
- Choose a primary key for every table, such as user_id or order_id.
- Add foreign keys to link tables and keep the data connected correctly.

Things to know

- Keep it simple at the start. Do not create too many tables too early.
- Avoid storing the same information in many places. This reduces mistakes.
- Use the right data types: text for names, date/time for dates, decimal for money.
- Decide what is required and what can be empty (NULL).
- Think about common questions people will ask, because that affects design and indexing.
- Add created_at and updated_at where history matters.

Easy checklist

1. What problem are we solving?
2. What are the main tables?
3. What columns do they need?
4. How are they connected?
5. What rules must always be true?

Small example

Customers → customer_id, name, email
Orders → order_id, customer_id, order_date
Order Items → order_item_id, order_id, product_id, quantity